Anwesha Sinha | GenAI + Full Stack Engineer | SDE-II | Institute Gold Medalist

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Summary

GenAI-focused Backend Engineer with 6+ years across fintech, ERP, and conversational AI. Expert in Java, Spring Boot, microservices, and secure, fault-tolerant systems; hands-on with Python, ReactJS, RAG, LangChain, FAISS, and AWS (EC2, S3, Lambda, RDS, DynamoDB, CloudWatch, IAM). Proven impact: automated financial workflows, reduced error rates with retry/DLQ, and scaled AI assistants with measurable gains in accuracy and efficiency.

Technical Skills

Programming: Java, Python, SQL, PL/SQL, JavaScript

Backend: Spring Boot, Microservices, REST APIs, Async processing, Schedulers

Frontend: ReactJS, HTML, CSS

AI/GenAI: RAG, LangChain, FAISS, LLM Integration (OpenAI, GPT-4), Prompt Engineering

Cloud: AWS: EC2, S3, Lambda, RDS, DynamoDB, CloudWatch, IAM, SQS/SNS

Data & Search: MongoDB, Oracle, Elasticsearch

Infra: Apache Kafka, Kong API Gateway, HashiCorp Vault, Docker, Kubernetes

Testing/CI-CD: JUnit, Mockito, Jenkins, GitHub Actions

Observability: Structured logging, ELK/Kibana, Metrics and dashboards

Experience

Airtel Payments Bank — **Gurgaon, India**

SDE-II (Manager) — *Aug 2023 – Present*

- Designed a resilient **retry engine** and high-availability microservices with **Spring Boot**, **Kafka**, and **Vault**, reducing failed transaction reprocess time by 60%.
- Built secure **REST APIs** for **APY Pension Scheme** with idempotency, masking, and audit logging; improved SLA adherence to 99.9%.
- Enhanced observability with structured logging, Kibana masking, and **CloudWatch**-style dashboards; decreased MTTR by 35%.
- Developed schedulers and automation for financial workflows (reconciliations, settlements), cutting manual interventions by 40%.
- Implemented DLQ patterns and backoff strategies; reduced message loss and improved throughput under peak load.

Oracle — **Bengaluru, India**

Software Engineer — *Aug 2021 – Aug 2023*

- Built scalable **ESS Jobs** and **Spring Boot APIs** in Fusion ERP, improving reporting efficiency by 40% and reducing batch runtime.
- Prototyped **AI-powered modules** within Joint Venture Cloud and Federal Financials: semantic search and intent pipelines on **Elasticsearch**.
- Optimized DB interactions (Oracle) using tuning, pagination, and batch writes; reduced query latency by 25%.

Yellow.ai — **Bengaluru, India**

AI Engineer — *May 2020 – Aug 2021*

- Developed NLP-driven bots and enterprise search using **Elasticsearch** and **MongoDB**; improved response accuracy by 15% via intent classification tuning.
- Integrated bots across **WhatsApp**, **Facebook**, and web; scaled multichannel deployments with robust webhook and auth flows.
- Built FAQ automation using LLMs + vector embeddings; reduced manual query load by 40%.
- Awarded **Spotlight Award** for Conversational AI innovation.

JK Lakshmi Cement Ltd. **Delhi, India**
Graduate Engineer Trainee *Jul 2019 – Mar 2020*

- Built dashboards in **Tableau** and Python automation for BI workflows; enabled data-driven decisions.
- Applied ML models to optimize resource usage and cut costs in production operations.

Projects

RAG Knowledge Assistant: Built a **Retrieval-Augmented Generation** system using **LangChain**, **FAISS**, and **OpenAI API** for financial knowledge search; improved answer relevance and reduced analyst lookup time.

Customer Support Chatbot: Developed **FAQ automation** with LLMs + vector embeddings, lowering manual query handling by 40%; added guardrails, prompt templates, and analytics.

End-to-End MCP Server: Engineered a scalable **Spring Boot MCP** server enabling secure integration with LLMs (GPT-4, LangChain, OpenAI Assistants API); standardized communication for enterprise AI agents and automated workflows.

Drool-worthy Decisions: Implemented **Drools Rule Engine** for dynamic decisioning in fintech, reducing manual rule changes by 70%; added auditability and versioning.

AWS Microservices Deployment: Containerized services with **Docker** and deployed to **AWS ECS/EC2**; used **ALB**, **Auto Scaling**, **CloudWatch** dashboards, and **IAM** policies for secure, monitored operations.

React Dashboard for GenAI: Built a **ReactJS** dashboard to manage datasets, embeddings, and inference pipelines for the RAG Assistant; implemented role-based access and usage metrics.

Education

JK Lakshmipat University **Jaipur, India**
B.Tech, Computer Science *2015 – 2019*
Gold Medalist CGPA: 9.2 75% Merit Scholarship

Certifications

Generative AI (O'Reilly); Apache Kafka (O'Reilly); Microservice Design (O'Reilly)

Oracle Cloud Infrastructure Foundations Associate (OCI)

Neo4J Certified Professional

Achievements

Spotlight Award – Conversational AI Innovation (2020)

Top 5% Smart India Hackathon (2017)

Gold Medalist, B.Tech (2019)

Publications

IJCA (2018): Comparative Study of Online Collaborative Tools for Marketing Effectiveness.

5th IGG Conf. (2017): Presented "Bag Tracking in IPO".

Keywords for ATS

AWS, EC2, S3, Lambda, RDS, DynamoDB, CloudWatch, IAM, SQS/SNS; Java, Spring Boot, Microservices, REST, Kafka, Docker, Kubernetes, Vault, Kong API Gateway; ReactJS, JavaScript; LangChain, FAISS, RAG, LLM, OpenAI, Elasticsearch, MongoDB, Oracle; JUnit, Mockito, Jenkins.